FEU Institute of Technology
COLLEGE OF ENGINEERING • COLLEGE OF COMPUTER STUDIES

Computer System and Architecture

Machine Problem

FINAL PROJECT

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I. Program Introduction

The Paws & Hearts Pet Adoption Center is a console-based NASM 32-bit Assembly program designed to manage pet adoption transactions. It allows users to view available dogs, cats, birds, and fish through an organized menu system. Each pet includes its name, breed, age, adoption fee, availability status, and ASCII art. Users can adopt pets, return them, and view the complete adoption history. The program also demonstrates the use of procedures, macros, file handling, and transactional processing in Assembly language.

II. Program Problem

Pet shelters may experience difficulties when adoption and return records are handled manually. Important information can be misplaced, duplicated, or incorrectly updated, making it difficult to know which pets are still available. Without an organized system, tracking adoption transactions can become time-consuming and unreliable.

III. Program Solution

The Paws & Hearts Pet Adoption Center Application Program provides a simple computerized solution for managing pet availability and transactions. It reads pet information from pets.txt and saves adoption and return records in separate files. Through its menu-driven interface, the program makes pet viewing, adoption, returning, and history checking easier and more organized.

IV. All Steps of Creating the Program

Step 1: Plan the Pet Adoption System

I first identified the main features that I wanted to include, such as viewing pets, adopting a pet, returning a pet, viewing history, and exiting the program. I decided to include four pet categories: dogs, cats, birds, and fish. This planning helped me organize the entire system before writing the Assembly code.

Step 2: Prepare the Program Files

I created the NASM source file named Final_Project.asm for the main program code. I also prepared pets.txt to store the pet names, breeds, ages, fees, and availability information. I allowed the program to create adoption_history.txt and returns.txt for storing transaction records.

Step 3: Define System Constants and Macros

I defined the Linux system call numbers needed for input, output, file opening, file reading, file writing, and file closing. I created macros such as PRINT for displaying messages and READ for accepting user input. Using macros helped me reduce repeated instructions and made the program more organized.

Step 4: Create the Program Data

I stored the titles, menu choices, prompts, labels, messages, pet names, breeds, ages, fees, and species in the data section. I added the ASCII art for the twelve pets and the pet adoption center. I carefully arranged the pet data so that each pet matched its correct information, status, and ASCII art.

Step 5: Build the Welcome Screen and Main Menu

I created a welcome screen that displays the Paws & Hearts Pet Adoption Center title and shelter ASCII art. I added the main menu options for viewing pets, adopting a pet, returning a pet, viewing adoption history, and exiting. I made the menu repeat continuously until the user selects the exit option.

Step 6: Implement Pet Category Selection

I created a procedure that displays the available pet categories. I allowed the user to choose dogs, cats, birds, or fish, after which the program displays three pets from the selected category. I included each pet's name, breed, age, adoption fee, availability status, and centered ASCII art.

Step 7: Implement the Adoption Procedure

I created an adoption procedure that asks the user to enter the name of a pet. I made the program search for the pet, verify that it exists, and check whether it is still available. After a successful adoption, I changed the pet's status and saved the transaction in adoption_history.txt.

Step 8: Implement the Return Procedure

I created a return procedure that asks the user to enter the name of the pet being returned. I made the program verify that the pet exists and is currently adopted. After a valid return, I changed the pet's status back to available and saved the transaction in returns.txt.

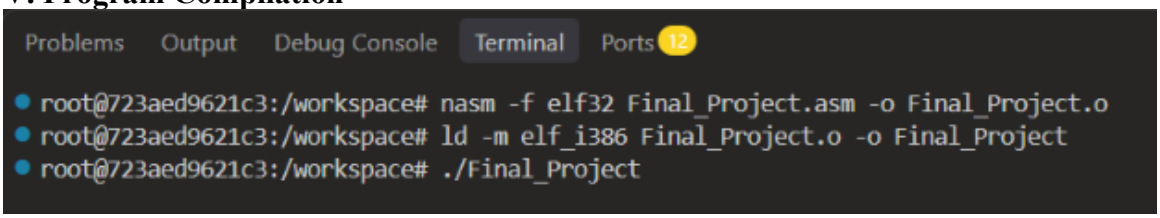
Step 9: Implement Adoption History and File Handling

I created file-handling procedures for opening, reading, writing, and closing files using Linux system calls. I made the history procedure read the adoption and return records from their files. I also organized the records into aligned table columns with borders so that the history would be easy to read.

Step 10: Compile, Test, and Demonstrate the Program

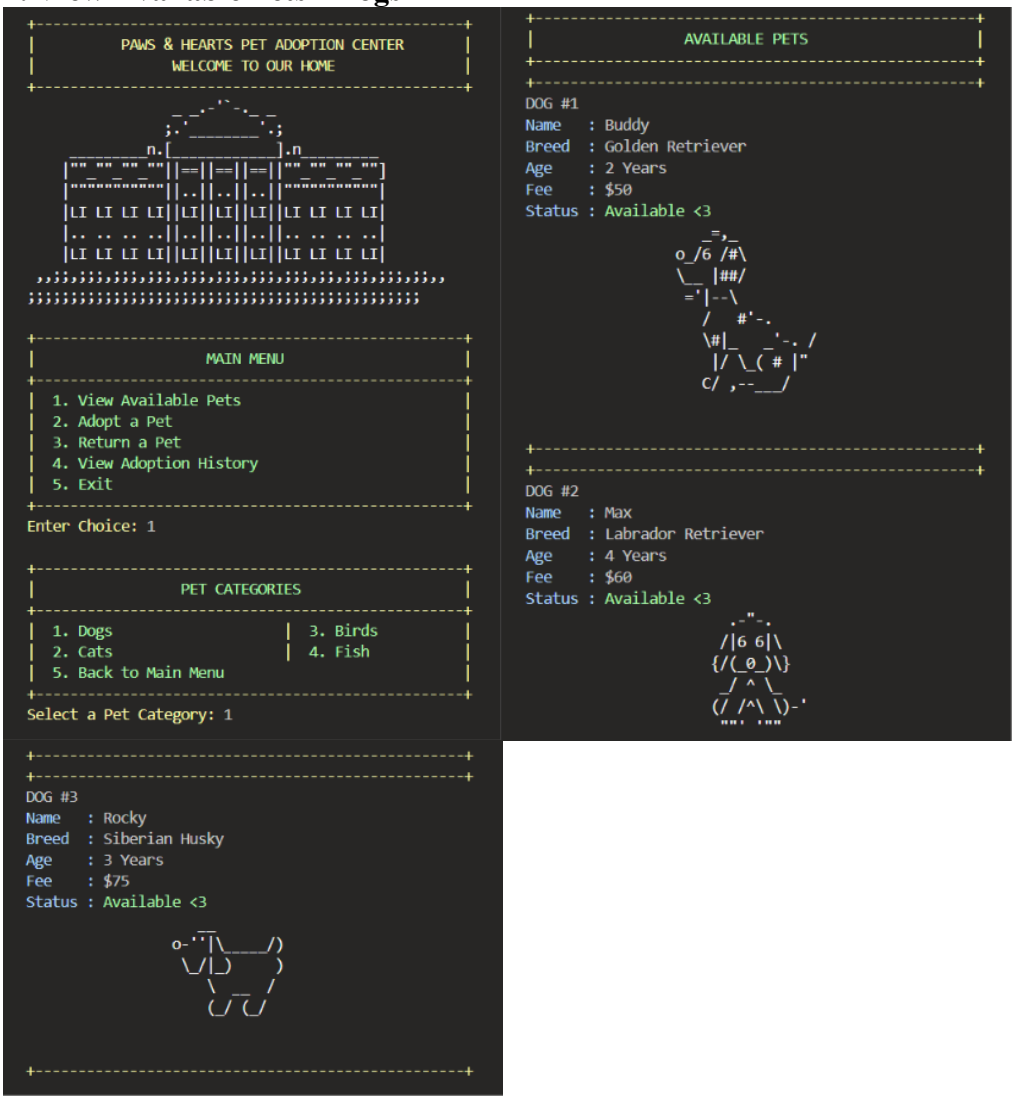
I compiled the source file using NASM and linked it as a 32-bit Linux executable. I tested every menu option, including viewing pets, adopting a pet, returning a pet, viewing history, entering invalid choices, and exiting the program. Finally, I demonstrated the welcome screen, pet ASCII art, file updates, adoption and return transactions, and the organized history display.

V. Program Compilation



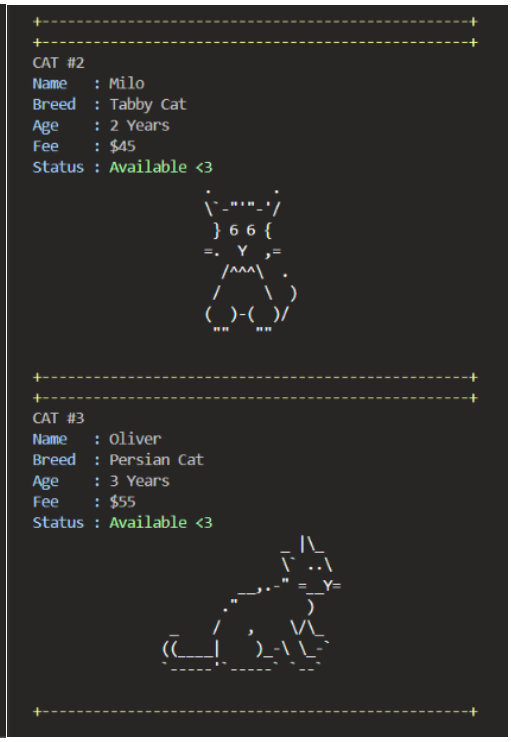
VI. Test Cases Outputs

1. View Available Pets - Dogs

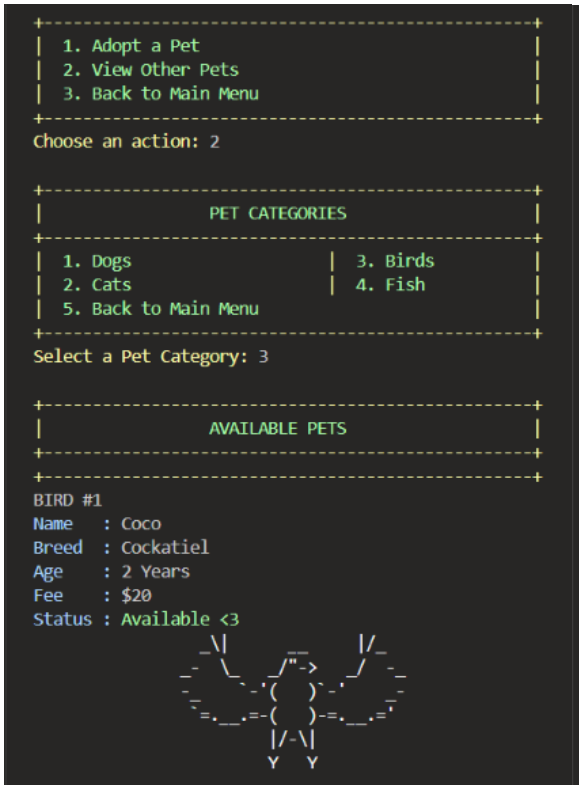


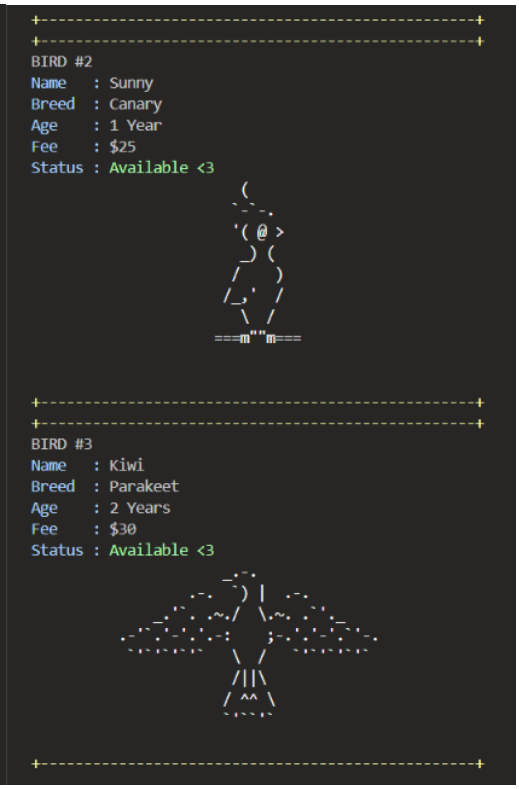
2. View Available Pets – Cats





3. View Available Pets – Birds





4. View Available Pets – Fishes



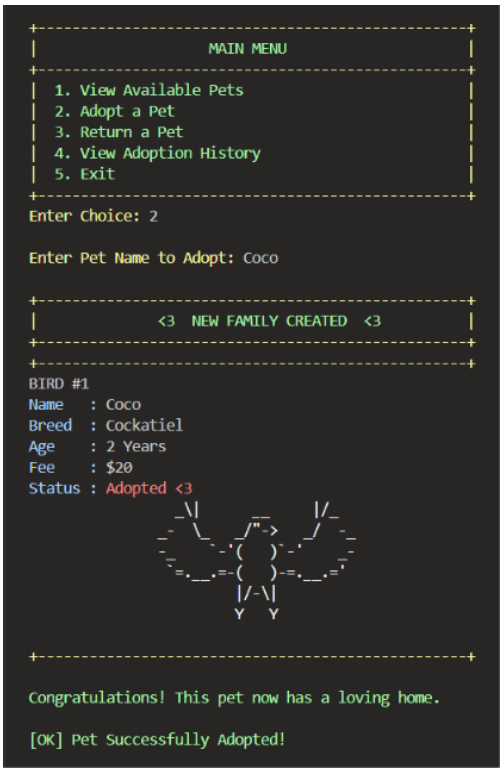
5. Adopt a Pet – Dog



6. Adopt a Pet – Cat



7. Adopt a Pet – Bird



8. Adopt a Pet – Fish



9. View Adoption History – Adopted Pets



10. Return a Pet

```
+--------------------------------------------------+
|                    MAIN MENU                     |
+--------------------------------------------------+
|  1. View Available Pets                          |
|  2. Adopt a Pet                                  |
|  3. Return a Pet                                 |
|  4. View Adoption History                        |
|  5. Exit                                         |
+--------------------------------------------------+
Enter Choice: 2

Enter Pet Name to Adopt: Nemo

+--------------------------------------------------+
|              <3  NEW FAMILY CREATED  <3          |
+--------------------------------------------------+
+--------------------------------------------------+
FISH #3
Name   : Nemo
Breed  : Clownfish
Age    : 1 Year
Fee    : $15
Status : Adopted <3
```



```
+--------------------------------------------------+

Congratulations! This pet now has a loving home.

[OK] Pet Successfully Adopted!

+--------------------------------------------------+
|                    MAIN MENU                     |
+--------------------------------------------------+
|  1. View Available Pets                          |
|  2. Adopt a Pet                                  |
|  3. Return a Pet                                 |
|  4. View Adoption History                        |
|  5. Exit                                         |
+--------------------------------------------------+
Enter Choice: 3

Enter Pet Name to Return: Nemo

[OK] Pet Successfully Returned!
```

11. View Adoption History – Adopted and Returned Pets

```
+--------------------------------------------------+
|                    MAIN MENU                     |
+--------------------------------------------------+
|  1. View Available Pets                          |
|  2. Adopt a Pet                                  |
|  3. Return a Pet                                 |
|  4. View Adoption History                        |
|  5. Exit                                         |
+--------------------------------------------------+
Enter Choice: 4

+--------------------------------------------------+
|                ADOPTION HISTORY                  |
+--------------------------------------------------+
+--------------------------------------------------+
|                ADOPTION RECORDS                  |
+--------------------------------------------------+
| STATUS   | PET NAME      | TYPE     | FEE        |
| ADOPTED  | Buddy         | Dog      | $50        |
| ADOPTED  | Luna          | Cat      | $40        |
| ADOPTED  | Coco          | Bird     | $20        |
| ADOPTED  | Bubbles       | Fish     | $10        |
| ADOPTED  | Nemo          | Fish     | $15        |
+--------------------------------------------------+
|                RETURN RECORDS                    |
+--------------------------------------------------+
| STATUS   | PET NAME      | TYPE     | FEE        |
| RETURNED | Nemo          | Fish     |            |
+--------------------------------------------------+
```

12. Adopt an Already Adopted Pet



13. Exit



VII. Source Code:

```
BITS 32
GLOBAL _start

%define SYS_EXIT 1
%define SYS_READ 3
%define SYS_WRITE 4
%define SYS_OPEN 5
%define SYS_CLOSE 6
```

```
%define STDIN 0
%define STDOUT 1
%define O_RDONLY 0
%define O_WRONLY 1
%define O_CREAT 64
%define O_APPEND 1024

%macro PRINT 1
    push %1
    call print_string
    add esp,4
%endmacro

%macro READ 2
    push %2
    push %1
    call read_line
    add esp,8
%endmacro

section .data
clear db 27,'[2J',27,'[H',0
reset db 27,'[0m',0
sparkles db
27,'[38;5;229m','★',27,'[38;5;218m','✿',27,'[38;5;222m','✦',27,'[38;5;183m','❀',27,'[38;5;217
m','♡',27,'[38;5;153m','✧',27,'[38;5;157m','❋',27,'[38;5;229m','☆',27,'[38;5;229m','★',27,'[3
8;5;218m','✿',27,'[38;5;222m','✦',27,'[38;5;183m','❀',27,'[38;5;217m','♡',27,'[38;5;153m','✧'
,27,'[38;5;157m','❋',27,'[38;5;229m','☆',27,'[38;5;229m','★',27,'[38;5;218m','✿',27,'[38;5;22
2m','✦',27,'[38;5;183m','❀',27,'[38;5;217m','♡',27,'[38;5;153m','✧',27,'[38;5;157m','❋',27,'[
38;5;229m','☆',27,'[38;5;229m','★',27,'[38;5;218m','✿',27,'[38;5;222m','✦',27,'[38;5;183m','
❀',27,'[38;5;217m','♡',27,'[38;5;153m','✧',27,'[38;5;157m','❋',27,'[38;5;229m','☆',27,'[38;5;
229m','★',27,'[38;5;218m','✿',27,'[38;5;222m','✦',27,'[38;5;183m','❀',27,'[38;5;217m','♡',27,
'[38;5;153m','✧',27,'[38;5;157m','❋',27,'[38;5;229m','☆',27,'[38;5;229m','★',27,'[38;5;218m',
'✿',27,'[38;5;222m','✦',27,'[38;5;183m','❀',27,'[38;5;217m','♡',27,'[38;5;153m','✧',27,'[38;5
;157m','❋',27,'[38;5;229m','☆',27,'[38;5;229m','★',27,'[38;5;218m','✿',27,'[38;5;222m','✦',2
7,'[38;5;183m','❀',27,'[38;5;217m','♡',27,'[38;5;153m','✧',27,'[38;5;157m','❋',27,'[38;5;229m
','☆',27,'[38;5;229m','★',27,'[38;5;218m','✿',27,'[38;5;222m','✦',27,'[38;5;183m','❀',27,'[3
8;5;217m','♡',27,'[38;5;153m','✧',27,'[38;5;157m','❋',27,'[38;5;229m','☆',10,27,'[38;5;229m',
'              <3  PAWS & HEARTS WELCOMES YOU  <3              ',10,27,'[0m',0
footer db
27,'[38;5;229m','★',27,'[38;5;218m','✿',27,'[38;5;222m','✦',27,'[38;5;183m','❀',27,'[38;5;217
```

```
m','♡',27,'[38;5;153m','✧',27,'[38;5;157m','❋',27,'[38;5;229m','☆',27,'[38;5;229m','★',27,'[3
8;5;218m','✿',27,'[38;5;222m','✦',27,'[38;5;183m','❀',27,'[38;5;217m','♡',27,'[38;5;153m','✧'
,27,'[38;5;157m','❋',27,'[38;5;229m','☆',27,'[38;5;229m','★',27,'[38;5;218m','✿',27,'[38;5;22
2m','✦',27,'[38;5;183m','❀',27,'[38;5;217m','♡',27,'[38;5;153m','✧',27,'[38;5;157m','❋',27,'[
38;5;229m','☆',27,'[38;5;229m','★',27,'[38;5;218m','✿',27,'[38;5;222m','✦',27,'[38;5;183m','
❀',27,'[38;5;217m','♡',27,'[38;5;153m','✧',27,'[38;5;157m','❋',27,'[38;5;229m','☆',27,'[38;5;
229m','★',27,'[38;5;218m','✿',27,'[38;5;222m','✦',27,'[38;5;183m','❀',27,'[38;5;217m','♡',27,
'[38;5;153m','✧',27,'[38;5;157m','❋',27,'[38;5;229m','☆',27,'[38;5;229m','★',27,'[38;5;218m',
'✿',27,'[38;5;222m','✦',27,'[38;5;183m','❀',27,'[38;5;217m','♡',27,'[38;5;153m','✧',27,'[38;5
;157m','❋',27,'[38;5;229m','☆',27,'[38;5;229m','★',27,'[38;5;218m','✿',27,'[38;5;222m','✦',2
7,'[38;5;183m','❀',27,'[38;5;217m','♡',27,'[38;5;153m','✧',27,'[38;5;157m','❋',27,'[38;5;229m
','☆',27,'[38;5;229m','★',27,'[38;5;218m','✿',27,'[38;5;222m','✦',27,'[38;5;183m','❀',27,'[3
8;5;217m','♡',27,'[38;5;153m','✧',27,'[38;5;157m','❋',27,'[38;5;229m','☆',10,27,'[0m',0
shelter db 27,'[38;5;231m','                _ _.-',39,'`-._
_',10,'               ;.',39,'________',39,'.;',10,'     _________n.[____________].n_________'
,10,'    |""_""_""_""||==||==||==||""_""_""_""',']',10,'    |"""""""""""||..||..||..||""""""""""
""|',10,'    |LI LI LI LI||LI||LI||LI||LI LI LI LI|',10,'    |.. .. .. ..||..||..||.. .. .. ..
..|',10,'    |LI LI LI LI||LI||LI||LI||LI LI LI LI|',10,'
,,;;;
```

```
Fish           ',27,'[38;5;229m','|',10,'|  ',27,'[38;5;157m','5. Back to Main
Menu                          ',27,'[38;5;229m','|',10,'+--------------------------------
-------------+',10,27,'[38;5;229m','Select a Pet Category: ',27,'[0m',0
actions db 10,27,'[38;5;229m','+-----------------------------------------------
+',10,'|  ',27,'[38;5;157m','1. Adopt a
Pet                                ',27,'[38;5;229m','|',10,'|  ',27,'[38;5;157m','2. View
Other Pets                           ',27,'[38;5;229m','|',10,'|  ',27,'[38;5;157m','3. Back
to Main Menu                         ',27,'[38;5;229m','|',10,'+--------------------------
----------------------+',10,27,'[38;5;229m','Choose an action: ',27,'[0m',0
available_title db 10,27,'[38;5;229m','+-----------------------------------------------
+',10,'|',27,'[38;5;157m','                AVAILABLE
PETS                   ',27,'[38;5;229m','|',10,'+--------------------------------------
-----+',10,27,'[0m',0
history_title db 10,27,'[38;5;229m','+-----------------------------------------------
+',10,'|',27,'[38;5;157m','               ADOPTION
HISTORY                 ',27,'[38;5;229m','|',10,'+--------------------------------------
-------+',10,27,'[0m',0
card_line db 27,'[38;5;229m','+-----------------------------------------------
+',27,'[0m',10,0

adopt_prompt db 10,27,'[38;5;229m','Enter Pet Name to Adopt: ',27,'[0m',0
return_prompt db 10,27,'[38;5;229m','Enter Pet Name to Return: ',27,'[0m',0
invalid_choice db 10,27,'[38;5;210m','[!] Invalid choice. Please try again.',27,'[0m',10,0
invalid_pet db 10,27,'[38;5;210m','[!] Pet not found. Use the exact pet name.',27,'[0m',10,0
already_adopted db 10,27,'[38;5;210m','[!] This pet is already adopted.',27,'[0m',10,0
not_adopted db 10,27,'[38;5;210m','[!] This pet is currently available.',27,'[0m',10,0
adoption_success db 10,27,'[38;5;157m','[OK] Pet Successfully Adopted!',27,'[0m',10,0
return_success db 10,27,'[38;5;157m','[OK] Pet Successfully Returned!',27,'[0m',10,0
file_error db 10,27,'[38;5;210m','[!] File operation failed.',27,'[0m',10,0
loaded db 10,27,'[38;5;157m','[OK] pets.txt loaded successfully.',27,'[0m',10,0
celebrate db 10,27,'[38;5;229m','+-----------------------------------------------
+',10,'|',27,'[38;5;157m','             <3  NEW FAMILY
CREATED  <3          ',27,'[38;5;229m','|',10,'+-------------------------------------------
---+',10,27,'[0m',0
congrats db 10,27,'[38;5;157m','Congratulations! This pet now has a loving home.',27,'[0m',10,0
no_records db 27,'[90m','(No records yet.)',27,'[0m',10,0
adoption_label db 27,'[38;5;229m','+-----------------------------------------------
+',10,'|',27,'[38;5;157m','               ADOPTION
RECORDS                ',27,'[38;5;229m','|',10,'+--------------------------------------
------+',10,27,'[0m',0
```

```
return_label db 27,'[38;5;229m','+-----------------------------------------------
+',10,'|',27,'[38;5;157m','                  RETURN
RECORDS                 ',27,'[38;5;229m','|',10,'+---------------------------------------
-------+',10,27,'[0m',0
history_columns db '|',27,'[38;5;157m',' STATUS    | PET NAME      | TYPE     |
FEE        ',27,'[38;5;229m','|',10,27,'[0m',0
row_start db '| ',0
history_tail db '       |',10,0
goodbye db 10,27,'[38;5;229m','+-----------------------------------------------
+',10,'|',27,'[38;5;157m','             THANK YOU FOR
VISITING               ',27,'[38;5;229m','|',10,'|',27,'[38;5;157m','          PAWS & HEARTS
PET CENTER                ',27,'[38;5;229m','|',10,'+---------------------------------------
--------+',10,27,'[0m',0
pastel_border db
27,'[38;5;229m','★',27,'[38;5;218m','✿',27,'[38;5;222m','✦',27,'[38;5;183m','❀',27,'[38;5;217
m','♡',27,'[38;5;153m','✧',27,'[38;5;157m','❋',27,'[38;5;229m','☆',27,'[38;5;229m','★',27,'[3
8;5;218m','✿',27,'[38;5;222m','✦',27,'[38;5;183m','❀',27,'[38;5;217m','♡',27,'[38;5;153m','✧'
,27,'[38;5;157m','❋',27,'[38;5;229m','☆',27,'[38;5;229m','★',27,'[38;5;218m','✿',27,'[38;5;22
2m','✦',27,'[38;5;183m','❀',27,'[38;5;217m','♡',27,'[38;5;153m','✧',27,'[38;5;157m','❋',27,'[
38;5;229m','☆',27,'[38;5;229m','★',27,'[38;5;218m','✿',27,'[38;5;222m','✦',27,'[38;5;183m','
❀',27,'[38;5;217m','♡',27,'[38;5;153m','✧',27,'[38;5;157m','❋',27,'[38;5;229m','☆',27,'[38;5;
229m','★',27,'[38;5;218m','✿',27,'[38;5;222m','✦',27,'[38;5;183m','❀',27,'[38;5;217m','♡',27,
'[38;5;153m','✧',27,'[38;5;157m','❋',27,'[38;5;229m','☆',27,'[38;5;229m','★',27,'[38;5;218m',
'✿',27,'[38;5;222m','✦',27,'[38;5;183m','❀',27,'[38;5;217m','♡',27,'[38;5;153m','✧',27,'[38;5
;157m','❋',27,'[38;5;229m','☆',27,'[38;5;229m','★',27,'[38;5;218m','✿',27,'[38;5;222m','✦',2
7,'[38;5;183m','❀',27,'[38;5;217m','♡',27,'[38;5;153m','✧',27,'[38;5;157m','❋',27,'[38;5;229m
','☆',27,'[38;5;229m','★',27,'[38;5;218m','✿',27,'[38;5;222m','✦',27,'[38;5;183m','❀',27,'[3
8;5;217m','♡',27,'[38;5;153m','✧',27,'[38;5;157m','❋',27,'[38;5;229m','☆',10,27,'[0m',0
main_title db 27,'[38;5;218m','                        ✿  MAIN
MENU  ✿                        ',27,'[0m',10,0
main_options db 27,'[38;5;157m','✿ 1. View Available
Pets                                  ✿',27,'[0m',10,0,27,'[38;5;157m','✿ 2. Adopt a
Pet                                        ✿',27,'[0m',10,0,27,'[38;5;157m','✿ 3.
Return a Pet                                    ✿',27,'[0m',10,0,27,'[38;5;157m','✿
4. View Adoption
History                                 ✿',27,'[0m',10,0,27,'[38;5;157m','✿ 5.
Exit                                          ✿',27,'[0m',10,0
category_title db 27,'[38;5;218m','                      ✿  PET
CATEGORIES  ✿                      ',27,'[0m',10,0
category_options db 27,'[38;5;157m','✿ 1. Dogs                3.
Birds                             ✿',27,'[0m',10,0,27,'[38;5;157m','✿ 2.
```

```
Cats                4. Fish                                ✿',27,'[0m',10,0,27,'[38;5;157m','✿
5. Back to Main Menu                                          ✿',27,'[0m',10,0
action_options db 27,'[38;5;157m','✿ 1. Adopt a
Pet                                                  ✿',27,'[0m',10,0,27,'[38;5;157m','✿ 2. View
Other Pets                                             ✿',27,'[0m',10,0,27,'[38;5;157m','✿ 3.
Back to Main Menu                                           ✿',27,'[0m',10,0
available_title2 db 27,'[38;5;218m','                        ✿  AVAILABLE
PETS  ✿                       ',27,'[0m',10,0
history_title2 db 27,'[38;5;218m','                        ✿  ADOPTION
HISTORY  ✿                      ',27,'[0m',10,0
celebrate2 db 27,'[38;5;229m','                    <3  NEW FAMILY
CREATED  <3                    ',27,'[0m',10,0
goodbye2 db 27,'[38;5;229m','                  <3  THANK YOU FOR
VISITING  <3                 ',27,'[0m',10,'                    PAWS & HEARTS PET
CENTER                    ',10,0

name_label db 27,'[38;5;153m','Name   : ',27,'[0m',0
breed_label db 27,'[38;5;153m','Breed  : ',27,'[0m',0
age_label db 27,'[38;5;153m','Age    : ',27,'[0m',0
fee_label db 27,'[38;5;153m','Fee    : ',27,'[0m',0
status_label db 27,'[38;5;153m','Status : ',27,'[0m',0
available db 27,'[38;5;157m','Available <3',27,'[0m',10,0
adopted db 27,'[38;5;210m','Adopted <3',27,'[0m',10,0
newline db 10,0
spacebuf db ' ',0
separator db ' | ',0
blank db 0
adopted_prefix db 'ADOPTED | ',0
returned_prefix db 'RETURNED | ',0

pets_file db 'pets.txt',0
adoption_file db 'adoption_history.txt',0
return_file db 'returns.txt',0

pet_names dd buddy,max,rocky,luna,milo,oliver,coco,sunny,kiwi,bubbles,goldie,nemo
pet_breeds dd
golden,lab,husky,siamese,tabby,persian,cockatiel,canary,parakeet,betta,goldfish,clownfish
pet_ages dd age2,age4,age3,age1,age2,age3,age2,age1,age2,age1,age2,age1
pet_fees dd fee50,fee60,fee75,fee40,fee45,fee55,fee20,fee25,fee30,fee10,fee12,fee15
pet_types dd dog,dog,dog,cat,cat,cat,bird,bird,bird,fish,fish,fish
pet_heads dd dog1,dog2,dog3,cat1,cat2,cat3,bird1,bird2,bird3,fish1,fish2,fish3
```

```
pet_art dd
exact_buddy,exact_max,exact_rocky,exact_luna,exact_milo,exact_oliver,exact_coco,exact_sunny,exa
ct_kiwi,exact_bubbles,exact_goldie,exact_nemo
pet_indents db 16,18,17,13,19,13,12,19,10,12,16,2
pet_categories db 1,1,1,2,2,2,3,3,3,4,4,4

buddy db 'Buddy',0
max db 'Max',0
rocky db 'Rocky',0
luna db 'Luna',0
milo db 'Milo',0
oliver db 'Oliver',0
coco db 'Coco',0
sunny db 'Sunny',0
kiwi db 'Kiwi',0
bubbles db 'Bubbles',0
goldie db 'Goldie',0
nemo db 'Nemo',0

golden db 'Golden Retriever',0
lab db 'Labrador Retriever',0
husky db 'Siberian Husky',0
siamese db 'Siamese Cat',0
tabby db 'Tabby Cat',0
persian db 'Persian Cat',0
cockatiel db 'Cockatiel',0
canary db 'Canary',0
parakeet db 'Parakeet',0
betta db 'Betta Fish',0
goldfish db 'Goldfish',0
clownfish db 'Clownfish',0

age1 db '1 Year',0
age2 db '2 Years',0
age3 db '3 Years',0
age4 db '4 Years',0
fee10 db '$10',0
fee12 db '$12',0
fee15 db '$15',0
fee20 db '$20',0
fee25 db '$25',0
```

```
fee30 db '$30',0
fee40 db '$40',0
fee45 db '$45',0
fee50 db '$50',0
fee55 db '$55',0
fee60 db '$60',0
fee75 db '$75',0
dog db 'Dog',0
cat db 'Cat',0
bird db 'Bird',0
fish db 'Fish',0

dog1 db 'DOG #1',10,0
dog2 db 'DOG #2',10,0
dog3 db 'DOG #3',10,0
cat1 db 'CAT #1',10,0
cat2 db 'CAT #2',10,0
cat3 db 'CAT #3',10,0
bird1 db 'BIRD #1',10,0
bird2 db 'BIRD #2',10,0
bird3 db 'BIRD #3',10,0
fish1 db 'FISH #1',10,0
fish2 db 'FISH #2',10,0
fish3 db 'FISH #3',10,0

big_dog1 db
27,'[38;5;229m',' .::=:..',10,' .=%@#*#+:
--=*%%%@*.',10,' .:#@-
.  ..+@=',10,' =@*.  -:
.%%:',10,' +@-  :@@@+:=  .*@#:',10,' -@-  .=
.*@@+  .:==.  .*=  .*@%.',10,'  =@:  :#.@-  *=%@@%. -
..%.  =@.',10,'  -
@*=.  .#:  .=#@+.  *..%:  .+=@.',10,'  +%.  -
%..#*.  :@@@@@.  *+  :@@.',10,'  +%.--.  :@.  =@@*..: -#  *@= -
@.',10,'  +@@*+  -%.  =@.  #: ..=@#*.',10,'  .+%@%==-.:%%  :@-
.  -%.. .@.  :+@:',10,'  +@+.*#-..:%%.-%@@@:* ...  =*#-. -
:%@*.',10,'  :%@*..=.  .#%:  :%*..*#%-@+=*=..:++:#@@-#%.',10,'  +@.%.  .@@@.
.%@*..  .=@@*-+#@=%@+  -=@.',10,'  :@: -  .#%..=@=:+*#..  .%. .%.  .#@.',10,'
:%%.  :@.*@=.  .=%:=.  .#=.-%.  .+=%.',10,'.@-%:  :@@.  .*@#-
:.  .%@:',10,'.@:.*. =:  -@:.=++##:  ==@*:+  .-:@@.',10,'
#%.  *=.*#.  =%=-  .@*=  =@#*#',10,'
```


big_dog2 db
big_dog3 db 27,'[38;5;229m','

```
..-  .*@@-.: .. .@--: :. .=.  .%#%@@#-.',10,':%@%#@=*.:+..=@@%:+: +. .@=%..*...--
=+%%=:.',10,'    .:*%@@@@%#:.*@@@@@@@@##@@@@@@%*+=:.',10,27,'[0m',10,0

cat_art1 db 27,'[38;5;218m','       /\\_/\\',10,'      /  o
o  \\',10,'     (    ^    )',10,'      \\  =  /',10,'       \\___/',10,'      /|   |\\',10,'
  /_|___|_\\',10,27,'[0m',10,0
cat_art2 db 27,'[38;5;218m','       /\\_/\\',10,'      ( =^.^=
)',10,'       )     (',10,'      (  ___  )',10,'       \\___/',10,'      /     \\',10,'     /__
_____\\',10,27,'[0m',10,0
cat_art3 db 27,'[38;5;218m','      |\\__/,|',10,'      |o . o |',10,'      |  >  |',10,'      |
\\_/ |',10,'       \\___/',10,'      /|   |\\',10,'     /_|___|_\\',10,27,'[0m',10,0
bird_art1 db 27,'[38;5;157m','         .-.',10,'        (o
o)',10,'        |=|=|',10,'       __| |__',10,'      /   V   \\',10,'     /___/
\\___\\',10,'        /  \\',10,27,'[0m',10,0
bird_art2 db 27,'[38;5;157m','         __',10,'      __/o
)',10,'     /  \\_/',10,'    /|   |',10,'     /   \\',10,'    /_____\\',10,'      /  \\',10,27,
'[0m',10,0
bird_art3 db 27,'[38;5;157m','        .--.',10,'       /  o \\',10,'      <   __
)',10,'       \\_/_/',10,'        /|',10,'       /_|_',10,'      /_____',10,27,'[0m',10,0
fish_art1 db
27,'[38;5;153m','         /\\',10,'    ____/  \\__',10,'  /     o     )><',10,'  \\____    __/'
,10,'       \\__/',10,'       /  \\',10,'      /____\\',10,27,'[0m',10,0
fish_art2 db 27,'[38;5;153m','       __',10,'      /  \\',10,'  ___/ o  )><',10,'
/   \\___/',10,' \\___/',10,'    /  \\',10,'   /____\\',10,27,'[0m',10,0
fish_art3 db 27,'[38;5;153m','       .-.',10,'     _(o
o)_',10,'    (  \\_/  )><',10,'     \\___/',10,'      /|\\',10,'     /_|_\\',10,'      /  \\',1
0,27,'[0m',10,0

exact_buddy db 27,'[38;5;231m','    _=,_',10,' o_/6 /#\',10,' \__ |##/',10,'  =',39,'|--
\',10,'    /   #',39,'-.',10,'    \#|_   _',39,'-. /',10,'     |/ \_( # |" ',10,'    C/ ,--
___/',10,10,27,'[0m',10,0
exact_max db 27,'[38;5;231m','     .-"-.',10,'    /|6 6|\',10,'   {/(_0_)\}',10,'    _/ ^
\_',10,'   (/ /^\ \)-',39,10,'    ""',39,' ',39,'"",10,27,'[0m',10,0
exact_rocky db 27,'[38;5;231m','   __',10,'o-',39,39,'|\_____/)',10,'
\_/|_)     )',10,'    \  __  /',10,'    (_/ (_/    ',10,10,27,'[0m',10,0
exact_luna db 27,'[38;5;231m',' _._     _,-',39,'""`-._',10,'(,-.`._,',39,'(       |\`-
/|',10,'    `-.-',39,' \ )-`( , o o)',10,'           `-    \`_`"',39,'-',10,27,'[0m',10,0
exact_milo db 27,'[38;5;231m',' .       .',10,' \`-"',39,'"-',39,'/',10,'  } 6 6 {',10,'
=.  Y  ,=',10,'   /^^^\  .',10,'  /     \  )',10,' (  )-(  )/',10,'  ""   "",10,27,'[0m',10,0
exact_oliver db 27,'[38;5;231m','               _ |\_',10,'               \`
..\',10,'          __,.-"
```

```
=__Y=',10,'          ."         )',10,'    _     /    ,     \/\_',10,'  ((____|     )_-\ \_-
`',10,'  `-----',39,'`-----` `--`',10,27,'[0m',10,0
exact_coco db 27,'[38;5;231m','      _\|       __      |/_',10,'   _-  \_   _/"->   _/  -
_',10,'   -_     `-',39,'(   )`-',39,'    _-',10,'    `=.__.=-(   )-
=.__.=',39,10,'            |/-\|',10,'            Y   Y',10,27,'[0m',10,0
exact_sunny db 27,'[38;5;231m','     (',10,'    `-`-.',10,'    ',39,'( @ >',10,'     _)
(',10,'    /    )',10,'   /_,',39,'  /',10,'     \  /',10,'  ===m""m===',10,10,27,'[0m',10,0
exact_kiwi db 27,'[38;5;231m','            _.-.',10,'        .-.  `) |  .-.',10,'    _.',39,'`.
.~./  \.~. .`',39,'._',10,'.-',39,'`.',39,'-',39,'.',39,'.-:    ;-.',39,'.',39,'-
',39,'.`',39,'-.',10,'
`',39,'`',39,'`',39,'`',39,'`   \  /   `',39,'`',39,'`',39,'`',39,'`',10,'            /||\',10
,'           / ^^ \',10,'            `',39,'``',39,'`',10,27,'[0m',10,0
exact_bubbles db 27,'[38;5;231m','          /"*._         _',10,'      .-*',39,'`    `*-._.-
',39,'/',10,'    < * ))     ,       (',10,'      `*-._`._(__.--*"`.\',10,27,'[0m',10,0
exact_goldie db 27,'[38;5;231m','|\   \\\\__     o',10,'| \_/    o \    o',10,'> _   ((
<_  oo',10,'| / \__+___/',10,'|/     |/',10,27,'[0m',10,0
exact_nemo db 27,'[38;5;231m','      /`·.',10,'     /,...,`:·',10,' ,·´  ,   `·.,·´)',10,':
© ):´;       ,  {',10,' `·., `·  ,·´\`·.)',10,'     `\\´´\,·´',10,10,27,'[0m',10,0

section .bss
input resb 128
buffer resb 8192
record resb 256
charbuf resb 2
fieldbuf resb 64
art_base_indent resd 1
art_indent_count resd 1
record_ptr resd 1
fd resd 1
category resb 1
status resb 12
history_len resd 1

section .text
_start:
    call init_files
    call read_pets
    PRINT clear
    PRINT welcome
    PRINT shelter
```

```
main_loop:
    call main_menu_proc
    cmp eax,1
    je view_option
    cmp eax,2
    je adopt_option
    cmp eax,3
    je return_option
    cmp eax,4
    je history_option
    cmp eax,5
    je exit_option
    PRINT invalid_choice
    jmp main_loop

view_option:
    call category_proc
    jmp main_loop
adopt_option:
    call adopt_pet
    jmp main_loop
return_option:
    call return_pet
    jmp main_loop
history_option:
    call view_history
    jmp main_loop
exit_option:
    PRINT clear
    PRINT goodbye
    PRINT shelter
    mov eax,SYS_EXIT
    xor ebx,ebx
    int 80h

main_menu_proc:
    PRINT main_menu
    READ input,128
    cmp byte [input+1],0
    jne bad_main
    mov al,[input]
```

```
    sub al,'0'
    cmp al,1
    jb bad_main
    cmp al,5
    ja bad_main
    movzx eax,al
    ret
bad_main:
    xor eax,eax
    ret

category_proc:
category_loop:
    PRINT category_menu
    READ input,128
    cmp byte [input+1],0
    jne bad_category
    mov al,[input]
    sub al,'0'
    cmp al,1
    jb bad_category
    cmp al,5
    ja bad_category
    cmp al,5
    je category_done
    mov [category],al
    call show_category
action_loop:
    PRINT actions
    READ input,128
    cmp byte [input+1],0
    jne bad_action
    cmp byte [input],'1'
    je action_adopt
    cmp byte [input],'2'
    je category_loop
    cmp byte [input],'3'
    je category_done
bad_action:
    PRINT invalid_choice
    jmp action_loop
```

```
action_adopt:
    call adopt_pet
    jmp action_loop
bad_category:
    PRINT invalid_choice
    jmp category_loop
category_done:
    ret

show_category:
    PRINT clear
    PRINT available_title
    xor esi,esi
show_loop:
    cmp esi,12
    je show_done
    mov al,[pet_categories+esi]
    cmp al,[category]
    jne show_next
    push esi
    call show_pet
    add esp,4
show_next:
    inc esi
    jmp show_loop
show_done:
    ret

show_pet:
    push ebp
    mov ebp,esp
    push esi
    mov esi,[ebp+8]
    PRINT card_line
    mov eax,[pet_heads+esi*4]
    PRINT eax
    PRINT name_label
    mov eax,[pet_names+esi*4]
    PRINT eax
    PRINT newline
    PRINT breed_label
```

```
    mov eax,[pet_breeds+esi*4]
    PRINT eax
    PRINT newline
    PRINT age_label
    mov eax,[pet_ages+esi*4]
    PRINT eax
    PRINT newline
    PRINT fee_label
    mov eax,[pet_fees+esi*4]
    PRINT eax
    PRINT newline
    PRINT status_label
    cmp byte [status+esi],1
    je show_adopted
    PRINT available
    jmp show_art
show_adopted:
    PRINT adopted
show_art:
    mov eax,[pet_art+esi*4]
    movzx edx,byte [pet_indents+esi]
    push edx
    push eax
    call print_art_indented
    add esp,8
    PRINT card_line
    pop esi
    pop ebp
    ret

adopt_pet:
    PRINT adopt_prompt
    READ input,128
    call find_pet
    cmp eax,-1
    je adopt_bad_name
    mov esi,eax
    cmp byte [status+esi],1
    je adopt_already
    mov byte [status+esi],1
    call build_adoption
```

```
    push record
    push adoption_file
    call append_file
    add esp,8
    cmp eax,1
    jne adopt_write_bad
    PRINT clear
    PRINT celebrate
    push esi
    call show_pet
    add esp,4
    PRINT congrats
    PRINT adoption_success
    ret
adopt_write_bad:
    mov byte [status+esi],0
    PRINT file_error
    ret
adopt_bad_name:
    PRINT invalid_pet
    ret
adopt_already:
    PRINT already_adopted
    ret

return_pet:
    PRINT return_prompt
    READ input,128
    call find_pet
    cmp eax,-1
    je return_bad_name
    mov esi,eax
    cmp byte [status+esi],0
    je return_not_adopted
    mov byte [status+esi],0
    call build_return
    push record
    push return_file
    call append_file
    add esp,8
    cmp eax,1
```

```
    jne return_write_bad
    PRINT return_success
    ret
return_write_bad:
    mov byte [status+esi],1
    PRINT file_error
    ret
return_bad_name:
    PRINT invalid_pet
    ret
return_not_adopted:
    PRINT not_adopted
    ret

find_pet:
    xor esi,esi
find_loop:
    cmp esi,12
    je find_none
    mov eax,[pet_names+esi*4]
    push input
    push eax
    call same_text
    add esp,8
    cmp eax,1
    je find_yes
    inc esi
    jmp find_loop
find_yes:
    mov eax,esi
    ret
find_none:
    mov eax,-1
    ret

same_text:
    push ebp
    mov ebp,esp
    push esi
    push edi
    mov esi,[ebp+8]
```

```
    mov edi,[ebp+12]
same_loop:
    mov al,[esi]
    cmp al,[edi]
    jne same_no
    cmp al,0
    je same_yes
    inc esi
    inc edi
    jmp same_loop
same_yes:
    mov eax,1
    jmp same_done
same_no:
    xor eax,eax
same_done:
    pop edi
    pop esi
    pop ebp
    ret

build_adoption:
    mov dword [record_ptr],record
    push adopted_prefix
    call add_text
    add esp,4
    mov eax,[pet_names+esi*4]
    push dword 13
    push eax
    call add_field
    add esp,8
    push separator
    call add_text
    add esp,4
    mov eax,[pet_types+esi*4]
    push dword 8
    push eax
    call add_field
    add esp,8
    push separator
    call add_text
```

```
    add esp,4
    mov eax,[pet_fees+esi*4]
    push eax
    call add_text
    add esp,4
    push newline
    call add_text
    add esp,4
    mov edi,[record_ptr]
    mov byte [edi],0
    ret

build_return:
    mov dword [record_ptr],record
    push returned_prefix
    call add_text
    add esp,4
    mov eax,[pet_names+esi*4]
    push dword 13
    push eax
    call add_field
    add esp,8
    push separator
    call add_text
    add esp,4
    mov eax,[pet_types+esi*4]
    push dword 8
    push eax
    call add_field
    add esp,8
    push separator
    call add_text
    add esp,4
    push dword 4
    push blank
    call add_field
    add esp,8
    push newline
    call add_text
    add esp,4
    mov edi,[record_ptr]
```

```
    mov byte [edi],0
    ret

add_text:
    push ebp
    mov ebp,esp
    push esi
    push edi
    mov esi,[ebp+8]
    mov edi,[record_ptr]
add_loop:
    mov al,[esi]
    cmp al,0
    je add_done
    mov [edi],al
    inc esi
    inc edi
    jmp add_loop
add_done:
    mov [record_ptr],edi
    pop edi
    pop esi
    pop ebp
    ret

add_field:
    push ebp
    mov ebp,esp
    push esi
    push edi
    mov esi,[ebp+8]
    mov ecx,[ebp+12]
    mov edi,[record_ptr]
field_copy:
    cmp byte [esi],0
    je field_pad
    mov al,[esi]
    mov [edi],al
    inc esi
    inc edi
    dec ecx
```

```
    cmp ecx,0
    je field_done
    jmp field_copy
field_pad:
    cmp ecx,0
    je field_done
    mov byte [edi],' '
    inc edi
    dec ecx
    jmp field_pad
field_done:
    mov [record_ptr],edi
    pop edi
    pop esi
    pop ebp
    ret

view_history:
    PRINT clear
    PRINT history_title
    PRINT adoption_label
    PRINT history_columns
    push adoption_file
    call display_history_file
    add esp,4
    PRINT return_label
    PRINT history_columns
    push return_file
    call display_history_file
    add esp,4
    PRINT card_line
    ret

init_files:
    push adoption_file
    call make_file
    add esp,4
    push return_file
    call make_file
    add esp,4
    ret
```

```
make_file:
    mov eax,SYS_OPEN
    mov ebx,[esp+4]
    mov ecx,O_WRONLY | O_CREAT | O_APPEND
    mov edx,0644o
    int 80h
    cmp eax,0
    jl make_done
    mov [fd],eax
    mov eax,SYS_CLOSE
    mov ebx,[fd]
    int 80h
make_done:
    ret

read_pets:
    mov eax,SYS_OPEN
    mov ebx,pets_file
    mov ecx,O_RDONLY
    xor edx,edx
    int 80h
    cmp eax,0
    jl read_bad
    mov [fd],eax
    mov eax,SYS_READ
    mov ebx,[fd]
    mov ecx,buffer
    mov edx,8192
    int 80h
    mov eax,SYS_CLOSE
    mov ebx,[fd]
    int 80h
    ret
read_bad:
    ret

append_file:
    mov eax,SYS_OPEN
    mov ebx,[esp+4]
    mov ecx,O_WRONLY | O_CREAT | O_APPEND
```

```
    mov edx,0644o
    int 80h
    cmp eax,0
    jl append_bad
    mov [fd],eax
    push dword [esp+8]
    call text_length
    add esp,4
    mov edx,eax
    mov eax,SYS_WRITE
    mov ebx,[fd]
    mov ecx,[esp+8]
    int 80h
    cmp eax,edx
    jne append_close_bad
    mov eax,SYS_CLOSE
    mov ebx,[fd]
    int 80h
    mov eax,1
    ret
append_close_bad:
    mov eax,SYS_CLOSE
    mov ebx,[fd]
    int 80h
append_bad:
    xor eax,eax
    ret

display_file:
    mov eax,SYS_OPEN
    mov ebx,[esp+4]
    mov ecx,O_RDONLY
    xor edx,edx
    int 80h
    cmp eax,0
    jl display_empty
    mov [fd],eax
    mov eax,SYS_READ
    mov ebx,[fd]
    mov ecx,buffer
    mov edx,8192
```

```
    int 80h
    cmp eax,0
    jle display_close
    mov edx,eax
    mov eax,SYS_WRITE
    mov ebx,STDOUT
    mov ecx,buffer
    int 80h
display_close:
    mov eax,SYS_CLOSE
    mov ebx,[fd]
    int 80h
    ret

display_history_file:
    mov eax,SYS_OPEN
    mov ebx,[esp+4]
    mov ecx,O_RDONLY
    xor edx,edx
    int 80h
    cmp eax,0
    jl history_empty
    mov [fd],eax
    mov eax,SYS_READ
    mov ebx,[fd]
    mov ecx,buffer
    mov edx,8192
    int 80h
    mov [history_len],eax
    mov eax,SYS_CLOSE
    mov ebx,[fd]
    int 80h
    cmp dword [history_len],0
    jle history_empty
    mov esi,buffer
history_line:
    mov edi,buffer
    add edi,[history_len]
    cmp esi,edi
    jae history_done
    cmp byte [esi],10
```

```
    je history_skip
    mov dword [record_ptr],record
    push row_start
    call add_text
    add esp,4
    push esi
    call parse_field
    add esp,4
    mov esi,eax
    push dword 8
    push fieldbuf
    call add_field
    add esp,8
    push separator
    call add_text
    add esp,4
    push esi
    call parse_field
    add esp,4
    mov esi,eax
    push dword 13
    push fieldbuf
    call add_field
    add esp,8
    push separator
    call add_text
    add esp,4
    push esi
    call parse_field
    add esp,4
    mov esi,eax
    push dword 8
    push fieldbuf
    call add_field
    add esp,8
    push separator
    call add_text
    add esp,4
    push esi
    call parse_field
    add esp,4
```

```
    mov esi,eax
    push dword 4
    push fieldbuf
    call add_field
    add esp,8
    push history_tail
    call add_text
    add esp,4
    mov edi,[record_ptr]
    mov byte [edi],0
    PRINT record
history_skip:
    cmp byte [esi],0
    je history_done
    inc esi
    jmp history_line
history_done:
    ret
history_empty:
    PRINT no_records
    ret

parse_field:
    push ebp
    mov ebp,esp
    push esi
    push edi
    mov esi,[ebp+8]
parse_leading:
    cmp byte [esi],' '
    jne parse_copy_start
    inc esi
    jmp parse_leading
parse_copy_start:
    mov edi,fieldbuf
parse_copy:
    mov al,[esi]
    cmp al,'|'
    je parse_finish
    cmp al,10
    je parse_finish
```

```
    cmp al,0
    je parse_finish
    mov [edi],al
    inc edi
    inc esi
    jmp parse_copy
parse_finish:
    cmp edi,fieldbuf
    je parse_terminate
    cmp byte [edi-1],' '
    jne parse_terminate
    dec edi
    jmp parse_finish
parse_terminate:
    mov byte [edi],0
    cmp byte [esi],'|'
    jne parse_return
    inc esi
parse_return:
    mov eax,esi
    pop edi
    pop esi
    pop ebp
    ret

display_empty:
    PRINT no_records
    ret

print_art_indented:
    push ebp
    mov ebp,esp
    push ebx
    push esi
    push edi
    mov esi,[ebp+8]
    mov eax,[ebp+12]
    mov [art_base_indent],eax
    mov edi,1
    mov ebx,1
art_loop:
```

```
    mov al,[esi]
    cmp al,0
    je art_done
    cmp edi,1
    jne art_output
    cmp ebx,1
    je art_add_indent
    cmp al,27
    je art_skip_indent
art_add_indent:
    mov eax,[art_base_indent]
    mov [art_indent_count],eax
art_indent_loop:
    cmp dword [art_indent_count],0
    je art_skip_indent
    PRINT spacebuf
    dec dword [art_indent_count]
    jmp art_indent_loop
art_skip_indent:
    mov al,[esi]
    mov edi,0
    xor ebx,ebx
art_output:
    mov [charbuf],al
    mov byte [charbuf+1],0
    PRINT charbuf
    cmp byte [charbuf],10
    jne art_next
    mov edi,1
art_next:
    inc esi
    jmp art_loop
art_done:
    pop edi
    pop esi
    pop ebx
    pop ebp
    ret

print_string:
    push ebp
```

```
    mov ebp,esp
    push esi
    mov esi,[ebp+8]
    xor edx,edx
print_len:
    cmp byte [esi+edx],0
    je print_now
    inc edx
    jmp print_len
print_now:
    mov eax,SYS_WRITE
    mov ebx,STDOUT
    mov ecx,esi
    int 80h
    pop esi
    pop ebp
    ret

read_line:
    push ebp
    mov ebp,esp
    push edi
    mov edi,[ebp+8]
    mov edx,[ebp+12]
    dec edx
    mov eax,SYS_READ
    mov ebx,STDIN
    mov ecx,edi
    int 80h
    cmp eax,0
    jle read_empty
    xor ecx,ecx
read_scan:
    cmp ecx,eax
    jae read_end
    cmp byte [edi+ecx],10
    je read_stop
    cmp byte [edi+ecx],13
    je read_stop
    inc ecx
    jmp read_scan
```

```
read_stop:
    mov byte [edi+ecx],0
    jmp read_done
read_end:
    mov byte [edi+eax],0
    jmp read_done
read_empty:
    mov byte [edi],0
read_done:
    pop edi
    pop ebp
    ret

text_length:
    push ebp
    mov ebp,esp
    mov ecx,[ebp+8]
    xor eax,eax
text_loop:
    cmp byte [ecx+eax],0
    je text_done
    inc eax
    jmp text_loop
text_done:
    pop ebp
    ret
```

VIII. Program Code Explanation

1. Input, Output

The program accepts user input through the keyboard, such as menu choices and pet names. It displays menus, pet information, transaction messages, and ASCII art on the terminal. The READ macro calls the read_line procedure to store the user's input in a buffer.

Source Code:

```
READ input,128

PRINT adopt_prompt
PRINT adoption_success
PRINT invalid_choice

read_line:
   push ebp
   mov ebp,esp
   push edi
```

```
    mov edi,[ebp+8]
    mov edx,[ebp+12]
    dec edx

    mov eax,SYS_READ
    mov ebx,STDIN
    mov ecx,edi
    int 80h

    mov byte [edi],0

    pop edi
    pop ebp
    ret
```

2. Menu and ASCII Art

The program uses a main menu to let the user select different functions. It also displays a pet adoption center design and individual ASCII art for each pet. The pet information and artwork are connected through arrays so that the correct art is displayed with the selected pet.

Source Code:

```
_start:
    call init_files
    call read_pets

    PRINT clear
    PRINT welcome
    PRINT shelter

main_loop:
    call main_menu_proc

    cmp eax,1
    je view_option
    cmp eax,2
    je adopt_option
    cmp eax,3
    je return_option
    cmp eax,4
    je history_option
    cmp eax,5
    je exit_option

    PRINT invalid_choice
    jmp main_loop

pet  art dd exact  buddy,exact  max,exact  rocky
        dd exact_luna,exact_milo,exact_oliver
        dd exact_coco,exact_sunny,exact_kiwi
        dd exact_bubbles,exact_goldie,exact_nemo
```

3. Procedures

I divided the program into procedures so that each part has a specific responsibility. Procedures are used for displaying pets, adopting, returning, reading files, writing files, and viewing history. This makes the program easier to understand, test, and maintain.

Source Code:

```
show_category:
    PRINT clear
    PRINT available_title

    xor esi,esi

show_loop:
    cmp esi,12
    je show_done

    mov al,[pet_categories+esi]
    cmp al,[category]
    jne show_next

    push esi
    call show_pet
    add esp,4

show_next:
    inc esi
    jmp show_loop

show_done:
    ret

show_pet:
    push ebp
    mov ebp,esp
    push esi

    mov esi,[ebp+8]

    PRINT card_line

    mov eax,[pet_names+esi*4]
    PRINT name_label
    PRINT eax

    mov eax,[pet_breeds+esi*4]
    PRINT breed_label
    PRINT eax

    mov eax,[pet_art+esi*4]
    push eax
    call print_art_indented
    add esp,8

    PRINT card_line

    pop esi
    pop ebp
    ret
```

4. Macros

I used macros to simplify repeated input and output instructions. The PRINT macro displays a string, while the READ macro receives input from the user. This makes the main program shorter and avoids repeating the same procedure calls many times.

Source Code:

```
%macro PRINT 1
    push %1
    call print_string
    add esp,4
%endmacro

%macro READ 2
    push %2
    push %1
    call read_line
    add esp,8
%endmacro

PRINT main_menu
READ input,128
```

5. File Handling: Reading and Writing

The program reads the pet catalog from pets.txt when it starts. It writes adoption records to adoption_history.txt and return records to returns.txt. Linux system calls are used to open, read, write, and close the files.

Source Code:

```
read_pets:
    mov eax,SYS_OPEN
    mov ebx,pets_file
    mov ecx,O_RDONLY
    xor edx,edx
    int 80h

    cmp eax,0
    jl read_bad

    mov [fd],eax

    mov eax,SYS_READ
    mov ebx,[fd]
    mov ecx,buffer
    mov edx,8192
    int 80h

    mov eax,SYS_CLOSE
    mov ebx,[fd]
    int 80h

read_bad:
    ret

append_file:
```

```
    mov eax,SYS_OPEN
    mov ebx,[esp+4]
    mov ecx,O_WRONLY | O_CREAT | O_APPEND
    mov edx,0644o
    int 80h

    mov [fd],eax

    mov eax,SYS_WRITE
    mov ebx,[fd]
    mov ecx,[esp+8]
    int 80h

    mov eax,SYS_CLOSE
    mov ebx,[fd]
    int 80h

    mov eax,1
    ret
```

6. Transaction Processing

The adoption procedure checks whether the entered pet name exists and whether the pet is still available. If the adoption is valid, the program changes the pet's status and creates an adoption record. The return procedure performs a similar process but changes the pet back to available.

Source Code:

```
adopt_pet:
    PRINT adopt_prompt
    READ input,128

    call find_pet
    cmp eax,-1
    je adopt_bad_name

    mov esi,eax
    cmp byte [status+esi],1
    je adopt_already

    mov byte [status+esi],1
    call build_adoption

    push record
    push adoption_file
    call append_file
    add esp,8

    PRINT adoption_success
    Ret

return_pet:
    PRINT return_prompt
    READ input,128

    call find_pet
    cmp eax,-1
```

```
    je return_bad_name

    mov esi,eax
    cmp byte [status+esi],0
    je return  not  adopted

    mov byte [status+esi],0
    call build_return

    push record
    push return_file
    call append_file
    add esp,8

    PRINT return_success
    ret
```

7. Pet Searching and Validation

The find_pet procedure searches through all twelve pet names. It compares the user's input with each stored name using the same_text procedure. This prevents invalid names from being adopted or returned and also prevents an already adopted pet from being adopted again.

Source Code:

```
find_pet:
    xor esi,esi

find_loop:
    cmp esi,12
    je find_none

    mov eax,[pet_names+esi*4]

    push input
    push eax
    call same_text
    add esp,8

    cmp eax,1
    je find_found

    inc esi
    jmp find_loop

find_found:
    mov eax,esi
    ret

find  none:
    mov eax,-1
    ret

same_text:
    push ebp
    mov ebp,esp
```

```
    mov esi,[ebp+8]
    mov edi,[ebp+12]

same_loop:
    mov al,[esi]
    cmp al,[edi]
    jne same_no

    cmp al,0
    je same_yes

    inc esi
    inc edi
    jmp same_loop

same_yes:
    mov eax,1
    ret

same_no:
    xor eax,eax
    ret
```

8. Adoption History Display

The program reads the adoption and return files and displays their contents in a structured table. Each field is given a fixed width so that names, types, fees, and statuses remain aligned. This makes the history easier to read during testing and demonstration.

Source Code:

```
view_history:
    PRINT clear
    PRINT history_title

    PRINT adoption_label
    PRINT history_columns

    push adoption_file
    call display_history_file
    add esp,4

    PRINT return_label
    PRINT history_columns

    push return_file
    call display_history_file
    add esp,4

    PRINT card  line
    Ret

history_columns db '| STATUS   | PET NAME      | TYPE     | FEE        |',10,0
row_start db '| ',0
history_tail db '       |',10,0
```

9. Pet Status Management

The program keeps a status value for every pet in the status array. A value of 0 represents an available pet, while a value of 1 represents an adopted pet. The status is checked whenever a pet is displayed, adopted, or returned.

Source Code:

```
status resb 12

cmp byte [status+esi],1
je show_adopted

PRINT available
jmp show_art

show_adopted:
   PRINT adopted

mov byte [status+esi],1

mov byte [status+esi],0
```

10. ANSI Colors and User Interface

I used ANSI escape sequences to make the application more attractive and easier to understand. Pastel yellow is used for borders and headings, pastel green is used for menu text and successful messages, and pastel blue is used for pet information labels. These colors help separate the different parts of the program while keeping the interface clean and readable.

Source Code:

```
main_menu db 27,'[38;5;229m',
        '+-------------------------------------------------+',10,
        '|',27,'[38;5;157m',
        '                 MAIN MENU                     ',
        27,'[38;5;229m','|',10,
        '+-------------------------------------------------+',10,0

adoption_success db 27,'[38;5;157m',
              '[OK] Pet Successfully Adopted!',
              27,'[0m',10,0

name_label db 27,'[38;5;153m',
         'Name   : ',
         27,'[0m',0
```